

PROGRESS REPORT ON THE ELECTRONICS SECTION OF THE INDUSTRY 4.0 SKILLING PROGRAM
UNDER LWERA ELECTRONICS AND SEMICONDUCTORS IN PARTNERSHIP WITH MAK UNIPOD
AND UNDP

Introduction

The industry 4.0 skilling program is an initiative by the science technology and innovation secretariat under the office of the president to equip graduates and the general public with the relevant skills needed to make Uganda become an active participant and producer of products under the 4th industrial revolution. The program is a partnership between STI, Lwera electronics and semiconductors, MAK unipod and UNDP.

This document, therefore is a brief report summarizing the progress in electronic section of the industry 4.0 skilling program given in chronological order from the week that the electronics sessions began

PROGRESS OVERVIEW

WEEK 1

25/06/25

- Introductory lessons on programming using C++ and C
- Introductory lessons on programming micro controllers using the Arduino IDE

26/06/25

- Introduction to Kicad EDA software and all its features and tools
- Introduction to supplementary tools like snapmagic for symbols and footprints for parts

WEEK 2

2/07/25

- Lessons on different communication protocols like I2C, SPI, UART, USART
- Simple demonstration on data sharing between ESP32 and computer via FTDI chip.

3/07/25

- Lessons on schematic creation footprint assignment and PCB layout and routing using kicad and snapmagic
- Design of a DC-DC buck converter using LM2596S chip.

WEEK 3

9/07/25

- Practice with tinkercad in developing and programming simple microcontroller circuits using the ESP32. Developed simple blinking led, and simple traffic light.

10/07/25

- Learnt how to read data sheets and Designed schematic of MPPT solar charge controller using its datasheet.
- Redesign and iterations to buck converter schematics and PCB

WEEK 4

16/07/25

- Lessons on using analog read and write functions for pwm control.
- Tested using LDR light sensor to control led brightness.

17/07/25

- Lessons on generation of manufacturing files from kicad like the gerber files, drill files, pick and place files and BOM files

WEEK 5

23/07/25

- C programming in context to microcontrollers and its implication to embedded systems beyond Arduino.

24/07/25

- Design of a heart rate sensor system based on the photoplethysmography (PPG) to serve as an introduction into the smart watch electronics

WEEK 6

30/07/25

- Introduction to IOT and how to network IOT devices to send data.
- Interfaced with platforms like thingspeak to send and receive data from an esp32 and MQTT software to open channels for communication.

31/07/25

- Research and design of block diagrams and component selection for the smart watch.
- Lessons on advanced topics in PCB design like impedance matching, differential pair routing.

GENERAL CHALLENGES FACED

- Lack of resources to practice on that has made it difficult to follow with the instructors has been and is the greatest problem faced.
- Limited practical time and chance to prototype our concepts, this is largely due to the lack of resources.